Lab Management Module

REGRESSION TEST CASE DOCUMENT

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Version: 1.0

Test Type: Regression Testing — Full Functional Coverage

Environment: Production-Ready (also applicable to LIVE/UAT)

Classification: QA Controlled Document

Regression Strategy & Scope

Regression Objective

Ensure that new builds, patches, or configuration changes to the EDC Lab Management module do not break previously working functionality. These test cases cover end-to-end functional paths, data integrity validations, permission boundary checks, inter-module impacts, and production-representative data scenarios.

What Makes These Regression Tests?

Each test case targets a specific behavior that has known risk of regression: permission boundary changes, variable setting locks post-deployment, audit trail continuity, flag state transitions, cross-form determinant evaluations, and export data fidelity under production-volume conditions.

Priority Definitions

P1-Critical	Data integrity or patient safety risk — must pass before release
P2-High	Core module functionality — failure blocks major workflows
P3-Medium	Supporting features — failure impacts specific but non-critical paths
P4-Low	Edge case / cosmetic — failure has minimal workflow impact

Regression Test Suite — Coverage Summary

Module Code	Regression Area	TC Count	Priority
REG-PRM	Lab Permissions — Boundary & Dependency Regression	12	P1-Critical
REG-LFM	Lab Form Integrity After Configuration Change	10	P1-Critical
REG-LVM	Lab Variable Setting Locks & PHI Guard Regression	10	P1-Critical
REG-RDV	Reference Date Variable Validation Regression	7	P2-High
REG-LAB	Lab CRUD & State Transition Regression	10	P1-Critical
REG-RRC	Reference Range — Data Integrity & Uniqueness Regression	12	P1-Critical
REG-DET	Range Determinant Logic — Cross-Form & Visit Regression	9	P1-Critical
REG-EFD	Effective Date — Validation & Bulk Override Regression	10	P2-High
REG-LCU	LOINC & Unit Data Persistence Regression	6	P2-High
REG-CDR	Copy & Deactivate Reference Range Regression	8	P2-High
REG-LDE	Data Entry — Lab Selection, Change & Persistence	11	P1-Critical
REG-OOR	Out-of-Range Flag — Trigger Accuracy & State Regression	12	P1-Critical
REG-FIP	Field Info Popup — Data Accuracy & Status Transition	7	P2-High
REG-UPL	Upload — Mapping Persistence & Flag Regression	9	P2-High
REG-EXP	Export — Column Fidelity & Flag State Regression	8	P2-High
REG-AUD	Audit Logs — Completeness & Accuracy Regression	10	P1-Critical
REG-DRL	Deactivate/Reactivate Lab — Flag & Data Persistence	8	P2-High
REG-MIG	Study Migration — Config Lock & Permission Regression	9	P1-Critical
REG-NEG	Negative Paths — Unsupported Feature Guards	10	P2-High
REG-PER	Performance & Concurrent Session Regression	6	P3-Medium
	TOTAL REGRESSION TEST CASES	184	

REG-PRM Lab Permissions — Boundary & Dependency Regression

Verifies permission boundaries hold after role edits, EDC Admin flag toggles, and cross-permission dependency rules.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-PRM-01	Permissions	P1-Critical	**Add/Edit Labs permission removal immediately revokes lab creation without session logout**	*User is logged in with Add/Edit Labs; Admin session open in separate browser*	1. As user, navigate to Manage → Lab Management — confirm + (Create) button is visible. 2. As Admin, remove Add/Edit Labs from the user's role and save. 3. As user, refresh the Lab Management page. 4. Attempt to click the Create Labs form and submit a new lab name.	After refresh: Create Labs form is no longer visible or is disabled. The + icon is absent. No new lab can be submitted. Existing labs remain readable if Read-Only Labs is still active.	N/E
REG-PRM-02	Permissions	P1-Critical	**Add/Edit Reference Ranges permission is auto-blocked when all other Lab permissions are removed**	*Role has Add/Edit Reference Ranges + Add/Edit Labs enabled; Add/Edit Labs is about to be removed*	1. Navigate to Configure → Roles & Permissions. 2. Uncheck Add/Edit Labs for the role (leaving only Add/Edit Reference Ranges). 3. Attempt to save the role. 4. Check whether Add/Edit Reference Ranges remains active.	System either auto-disables Add/Edit Reference Ranges or shows validation: 'Add/Edit Reference Ranges requires at least one other Lab permission'. Saving is blocked or the permission is cleared automatically.	N/E
REG-PRM-03	Permissions	P1-Critical	**Re-enabling Labs under EDC Admin restores the Labs permission section and Manage menu item**	*Labs was disabled under EDC Admin → Enable; all Lab permissions hidden*	1. Confirm Labs section absent from Roles & Permissions and Manage menu. 2. Enable Labs under EDC Admin → Enable → save. 3. Navigate to Configure → Roles & Permissions → any role → PERMISSIONS tab. 4. Navigate to Manage menu.	Labs section reappears under PERMISSIONS with all four sub-permissions. 'Lab Management' reappears in the Manage menu. No page reload beyond standard navigation is required.	N/E
REG-PRM-04	Permissions	P2-High	**Form-level Read Only permission under Add/Edit Reference Ranges persists after role re-save**	*A role has Add/Edit Reference Ranges with Form_A set to Read Only*	1. Re-open the role in Configure → Roles & Permissions. 2. Make an unrelated permission change (e.g., toggle a non-Lab permission). 3. Save the role. 4. Log in as a user with this role → navigate to Lab Management → reference range for Form_A.	Form_A reference ranges remain Read Only. The unrelated permission change did not reset form-level overrides. Data integrity of form-level settings is preserved across role saves.	N/E
REG-PRM-05	Permissions	P1-Critical	**Deactivate Labs permission cannot deactivate a lab when Add/Edit Labs is absent**	*Role has only Deactivate Labs permission, no Add/Edit Labs*	1. Log in as user with Deactivate Labs only. 2. Navigate to Manage → Lab Management. 3. Click three-dot icon next to an active lab. 4. Click 'Deactivate Lab'.	Lab is deactivated successfully because the user has the specific Deactivate Labs permission. However, the DETAILS fields remain uneditable. The + (Create) button is absent. Deactivation action is scoped correctly.	N/E
REG-PRM-06	Permissions	P2-High	**Study Admin without Lab permissions cannot access Lab Management even in LIVE**	*Study Admin role has no Lab permissions; LIVE study is the active environment*	1. Log in as Study Admin in the LIVE environment. 2. Navigate to Manage menu. 3. Check for 'Lab Management' option.	'Lab Management' is absent from the Manage menu. Study Admin cannot access or view any lab data without explicit Lab permission grants. Environment (LIVE vs DEV) does not bypass permission enforcement.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-PRM-07	Permissions	P2-High	**No Access / Hide Form under Reference Ranges completely hides the form from that role**	*Role has Add/Edit Reference Ranges with Form_B set to 'No Access Hide Form'*	1. Log in as a user with this role. 2. Navigate to Manage → Lab Management. 3. Attempt to find or navigate to reference ranges for Form_B.	Form_B is completely hidden. It does not appear in the LAB TESTS form checklist or in the Manage Ranges view. The form is treated as non-existent for this user's session.	N/E
REG-PRM-08	Permissions	P1-Critical	**Downgrading from Add/Edit Labs to Read-Only Labs does not expose orphaned edit controls**	*User previously had Add/Edit Labs; role was changed to Read-Only Labs*	1. User opens Lab Management after role downgrade. 2. Inspect DETAILS tab, LAB TESTS tab, Manage Ranges. 3. Attempt to interact with any field or button.	All write controls (Save, Create, Manage Ranges edit icons) are absent or disabled. No stale UI element allows data submission. The downgrade is reflected completely in the UI without leftover edit capabilities.	N/E
REG-PRM-09	Permissions	P2-High	**Read & Write form permission under Reference Ranges allows full range edit including effective dates**	*Role has Add/Edit Reference Ranges; Form_C is set to Read & Write*	1. Log in as the user → navigate to Manage → Lab Management → select a lab. 2. Check Form_C → Manage Ranges → click edit icon for a variable. 3. Edit lower limit, upper limit, and effective dates. 4. Save.	All range fields including lower limit, upper limit, and effective dates are editable and saveable for Form_C. No read-only restrictions. Save is confirmed with updated values visible on revisit.	N/E
REG-PRM-10	Permissions	P1-Critical	**Permission-gated Lab Management endpoint cannot be accessed by direct URL manipulation**	*User role has no Lab permissions; knows the Lab Management URL path*	1. Log in as a user with no Lab permissions. 2. Manually type the Lab Management URL in the browser address bar. 3. Observe system response.	System redirects to an access-denied page or returns to the dashboard with an appropriate error. Direct URL access does not bypass permission enforcement.	N/E
REG-PRM-11	Permissions	P2-High	**Concurrent role edit by two admins does not result in permission corruption**	*Two Admin sessions open simultaneously; both viewing the same role's permissions*	1. Admin A enables Add/Edit Labs for Role_QA. 2. Simultaneously, Admin B enables Read-Only Labs for the same Role_QA. 3. Both admins click Save. 4. Verify the final state of Role_QA permissions.	System applies last-write-wins or throws a conflict error. The final permission state is coherent — no partial or contradictory permission combination results. Audit reflects which admin's save persisted.	N/E
REG-PRM-12	Permissions	P2-High	**Reference Range Audit Log respects permission — read-only user cannot download it**	*User has Read-Only Labs permission only*	1. Log in as Read-Only user. 2. Navigate to Manage → Lab Management. 3. Click three-dot icon next to Save. 4. Check if 'Reference Range Audit Log' is available for download.	Read-Only user can view labs but the audit log download option is either absent or produces a permission-denied message. Audit log access is gated to users with appropriate permissions.	N/E

REG-LFM Lab Form Integrity After Configuration Change

Verifies Lab form assignments, visit bindings, and restrictions hold correctly after form edits and deployments.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LFM-01	Lab Forms	P1-Critical	**Lab form retains Lab use type after editing the form name**	*Lab form 'Hematology Panel' exists with form use = Lab*	1. Navigate to Configure → Forms → Hematology Panel. 2. Rename the form to 'Hematology Panel v2'. 3. Save the form. 4. Navigate to Manage → Lab Management → any lab → LAB TESTS tab.	Renamed form 'Hematology Panel v2' still appears in the LAB TESTS form list with its Lab use type intact. No re-assignment of form use type is required after a name change.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LFM-02	Lab Forms	P1-Critical	**Removing a visit from a Lab form does not deactivate reference ranges for remaining visits**	*Lab form 'CBC Panel' is assigned to Screening, Week 4, Week 8; reference ranges configured*	1. Navigate to Configure → Forms → CBC Panel. 2. Remove 'Week 8' visit from the form. 3. Save. 4. Navigate to Lab Management → Manage Ranges for CBC Panel → check reference ranges.	Reference ranges for Screening and Week 4 remain active and unchanged. Only Week 8-specific determinant visit configurations should be reviewed. Ranges are not mass-deactivated by the visit removal.	N/E
REG-LFM-03	Lab Forms	P1-Critical	**Standard form converted to Lab form use type appears in Lab Management for range configuration**	*An existing standard form 'Lipid Panel' has no Lab use type assigned*	1. Navigate to Configure → Forms → Lipid Panel → edit. 2. Change form use from --N/A-- to 'Lab'. 3. Navigate to Manage → Lab Management → any lab → LAB TESTS tab.	'Lipid Panel' appears in the LAB TESTS checklist. Upon selection and clicking Manage Ranges, Lab Test Result variables from Lipid Panel are listed for range setup.	N/E
REG-LFM-04	Lab Forms	P2-High	**Copying a Lab form preserves the Lab use type in the copied version**	*Lab form 'Metabolic Panel' exists with Lab use type*	1. Navigate to Configure → Forms → Add / Import / Copy a Form → Copy. 2. Select 'Metabolic Panel' as source. 3. Complete the copy. 4. Check the copied form's use type.	Copied form inherits Lab use type from source. It appears in the LAB TESTS tab in Lab Management without requiring manual re-assignment of the Lab use type.	N/E
REG-LFM-05	Lab Forms	P1-Critical	**ePRO form retains its restriction from being assigned Lab use type after system update**	*An ePRO form 'Patient Diary' exists in the study*	1. Navigate to Configure → Forms → Patient Diary. 2. Attempt to change form use to 'Lab' via the form editor or import. 3. Observe validation behavior.	System continues to block Lab use type assignment for ePRO forms post-update. Restriction is enforced consistently. An appropriate error or disabled state prevents the assignment.	N/E
REG-LFM-06	Lab Forms	P2-High	**Lab form subject data is not lost when the form is edited (visit or label change)**	*Subject S-100 has Screening visit data in CBC Panel; form label is being updated*	1. Navigate to Configure → Forms → CBC Panel. 2. Update the form label from 'CBC Panel' to 'Complete Blood Count'. 3. Save. 4. Navigate to S-100's Screening CBC Panel data.	S-100's existing data entries are fully preserved. The form displays under the new label 'Complete Blood Count' but all entered values, selected labs, and out-of-range flags remain intact.	N/E
REG-LFM-07	Lab Forms	P2-High	**Lab form with DDC data entry does not show Out-of-Range flags in DDC after patch**	*A Lab variable 'INR' has a reference range 1.0–1.5; DDC data entry for subject S-101 has INR=2.0*	1. Open the Lab form for S-101 in DDC mode. 2. Enter INR = 2.0 and save via DDC. 3. Switch to EDC view for the same form and subject.	DDC view: No out-of-range flag appears. EDC view: Out-of-range flag is present. This DDC restriction is a known design behavior — regression confirms it has not been inadvertently fixed (removing flags from EDC) or broken (adding flags to DDC).	N/E
REG-LFM-08	Lab Forms	P1-Critical	**Lab form reference ranges are not disrupted when a non-Lab variable is added to the same form**	*CBC Panel has WBC Count (Lab Test Result) with active range; a new text variable 'Sample Note' is added*	1. Add a standard Text variable 'Sample Note' to CBC Panel. 2. Save the form. 3. Navigate to Lab Management → select CBC Panel → Manage Ranges.	WBC Count reference ranges are fully intact after adding 'Sample Note'. The new text variable does not appear in Manage Ranges (it is not a Lab Test Result). No range disruption.	N/E
REG-LFM-09	Lab Forms	P2-High	**Importing a new version of an existing Lab form does not duplicate or overwrite active reference ranges**	*CBC Panel has three active reference ranges; a new version of CBC Panel is being imported*	1. Import a new version of CBC Panel (same variable names, updated labels). 2. Navigate to Lab Management → select the lab → LAB TESTS → CBC Panel → Manage Ranges.	Existing three reference ranges are preserved. Import does not create duplicates or overwrite existing ranges. If the import creates a new form, ranges are scoped to the original form only.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LFM-10	Lab Forms	P2-High	**Deleting a Lab variable deactivates its reference ranges without affecting other variables' ranges**	*CBC Panel has WBC Count and Hemoglobin both as Lab Test Result with independent ranges*	1. Delete 'WBC Count' variable from CBC Panel. 2. Navigate to Lab Management → Manage Ranges for CBC Panel.	All WBC Count reference ranges are deactivated (shown as inactive). Hemoglobin ranges are completely unaffected — still active and editable. Deactivation is scoped to the deleted variable only.	N/E

REG-LVM Lab Variable Setting Locks & PHI Guard Regression

Confirms variable Lab settings cannot be changed post-deployment and PHI restrictions are enforced consistently.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LVM-01	Lab Variables	P1-Critical	**Lab Test Result setting remains locked after the variable is used in Lab Management**	*WBC Count is set as Lab Test Result and has been referenced in at least one reference range*	1. Navigate to Configure → Forms → CBC Panel → WBC Count → Variable Editor. 2. Attempt to change 'Lab Test Result' to 'None' or 'Range Determinant'. 3. Click Save.	System blocks the change: 'This variable has been used in Lab Management. Configuration settings cannot be changed.' The Lab Test Result setting persists. Save is denied.	N/E
REG-LVM-02	Lab Variables	P1-Critical	**Range Determinant setting is locked after the variable is referenced in a reference range**	*Gender (Categorical Select) is set as Range Determinant and used in a reference range determinant condition*	1. Navigate to the Gender variable's Variable Editor. 2. Attempt to change the Lab Range setting from Range Determinant to None. 3. Save.	Setting change is blocked: 'This variable is referenced in a Lab Management reference range. Settings cannot be modified.' Gender remains as Range Determinant.	N/E
REG-LVM-03	Lab Variables	P1-Critical	**PHI flag cannot be enabled on a Lab Test Result variable even after system patches**	*WBC Count has Lab Test Result setting; a system patch was recently applied*	1. Open WBC Count in Variable Editor. 2. Verify Lab Test Result is still set. 3. Attempt to check the PHI checkbox. 4. Click Save.	PHI checkbox is either disabled when Lab Test Result is active, or Save is blocked with: 'Lab Test Result variables cannot be marked as PHI.' Regression confirms the patch did not inadvertently enable this combination.	N/E
REG-LVM-04	Lab Variables	P1-Critical	**Range Determinant variable PHI restriction is enforced across all variable types**	*Categorical Select 'Pregnancy Status' is set as Range Determinant*	1. Open Pregnancy Status in Variable Editor. 2. Attempt to enable PHI. 3. Save.	PHI cannot be set on any Range Determinant variable regardless of type. Blocked with appropriate error. Regression confirms PHI guard is not type-specific — it covers Number, Categorical Select, Yes/No, and Calculated types equally.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LVM-05	Lab Variables	P1-Critical	**Calculated variable (Age) retains Range Determinant setting after study re-open or session expiry**	*Calculated variable 'Age' is set as Range Determinant and referenced in a reference range*	1. Log out and log back in. 2. Navigate to the Age variable's Variable Editor. 3. Verify the Lab Range setting.	Range Determinant setting persists across sessions. No setting drift occurs after session expiry or page refresh.	N/E
REG-LVM-06	Lab Variables	P2-High	**Lab Test Display Label change updates the display in Lab Management without altering the variable's data**	*WBC Count variable has Lab Test Display Label 'WBC (cells/uL)'; label is being updated*	1. Open WBC Count Variable Editor. 2. Change Lab Test Display Label to 'White Blood Cell Count (K/uL)'. 3. Save. 4. Navigate to Lab Management → Manage Ranges.	Manage Ranges view shows 'White Blood Cell Count (K/uL)' as the display label. Existing reference ranges are intact. Subject data and flags are unaffected by the label change.	N/E
REG-LVM-07	Lab Variables	P1-Critical	**Numeric variable inside a grid retains Lab Test Result setting after grid structure change**	*A pre-filled grid in CBC Panel has a numeric variable 'Neutrophils' set as Lab Test Result*	1. Modify the grid (e.g., add a new column 'Basophils'). 2. Save the form. 3. Open Neutrophils in Variable Editor. 4. Navigate to Lab Management → Manage Ranges.	Neutrophils retains Lab Test Result setting after the grid structural change. It remains visible in Manage Ranges. No reset of its lab setting due to grid edit.	N/E
REG-LVM-08	Lab Variables	P2-High	**Deleting and recreating a Lab Test Result variable creates fresh reference range slots**	*WBC Count was deleted (ranges deactivated); new 'WBC_v2' is created as Lab Test Result*	1. Confirm WBC Count ranges are deactivated in Lab Management. 2. Create new variable 'WBC_v2' on CBC Panel as Lab Test Result. 3. Navigate to Lab Management → Manage Ranges.	'WBC_v2' appears in Manage Ranges with no pre-existing ranges. User can configure new ranges starting from default 0–0. Deactivated WBC Count ranges are separate and not re-linked to WBC_v2.	N/E
REG-LVM-09	Lab Variables	P2-High	**Lab Test Display Label is included in Export Labs and Data Export correctly after a label update**	*WBC Count display label was changed to 'White Blood Cell Count (K/uL)'*	1. Export Labs and open the Excel file. 2. Locate WBC Count entries. 3. Export data with Include Labs and inspect the variable column header.	Export uses the updated display label 'White Blood Cell Count (K/uL)' in relevant columns. Label changes propagate correctly to all export types.	N/E
REG-LVM-10	Lab Variables	P1-Critical	**Only one Lab setting per variable is enforced via API-level validation, not just UI**	*Developer attempts to set both Lab Test Result and Range Determinant via an API call on the same variable*	1. Simulate or test an API/backend call that attempts to assign both Lab Test Result and Range Determinant to the same variable. 2. Observe the server response.	Server rejects the request with an appropriate error. Backend validation mirrors UI validation. Setting both simultaneously is blocked at every layer — not just the UI.	N/E

REG-RDV Reference Date Variable Validation Regression

Validates Reference Date variable restrictions hold after study changes, including PHI, grid, and partial-date guards.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-RDV-01	Ref Date	P2-High	**Reference Date variable enforces full date requirement after study re-open**	*Lab form has a Date variable 'Sample Date' set as Reference Date with Day/Month/Year required*	1. Re-open the study after a deployment. 2. Open 'Sample Date' Variable Editor. 3. Verify Day Required, Month Required, Year Required are all still checked. 4. Attempt to uncheck 'Day Required' and save.	All three date fields remain required. Unchecking any one is blocked: 'Reference Date requires a complete date — Day, Month, and Year must all be required.' Settings are stable across deployments.	N/E
REG-RDV-02	Ref Date	P1-Critical	**Reference Date variable on Lab Form A is not visible in the Effective Dates panel for Lab Form B**	*Lab Form A has 'Lab Collection Date' as Reference Date; Lab Form B is a separate Lab form*	1. Navigate to Lab Management → select a lab → LAB TESTS → check Lab Form B → Manage Ranges. 2. Click on a reference range → go to Effective Dates section.	The Effective Dates section for Form B ranges does not offer 'Lab Collection Date' from Form A. Only Reference Date variables on Form B itself are available. Cross-form Reference Date linkage is not permitted.	N/E
REG-RDV-03	Ref Date	P1-Critical	**PHI flag is blocked on Reference Date variable after a system patch**	*Lab form has 'Sample Date' set as Reference Date; system patch was recently applied*	1. Open 'Sample Date' in Variable Editor. 2. Verify Reference Date setting is still active. 3. Attempt to enable PHI. 4. Save.	PHI remains blocked for Reference Date variables. Patch did not inadvertently enable this combination. Error or disabled state prevents the PHI assignment.	N/E
REG-RDV-04	Ref Date	P2-High	**Reference Date variable cannot be dragged into a grid in the form editor**	*A Lab form with a pre-filled grid exists; 'Sample Date' is a Reference Date variable outside the grid*	1. In the form editor, attempt to drag 'Sample Date' into the grid. 2. Observe system behavior.	System blocks the placement. 'Sample Date' cannot be moved into a grid. Error: 'Effective date variables cannot be placed in a grid.' Restriction is enforced via UI drag-and-drop constraints.	N/E
REG-RDV-05	Ref Date	P2-High	**Date and Time variable set as Reference Date retains its type setting after form save**	*Lab form 'Biochemistry Panel' has 'Sample Timestamp' (Date and Time) set as Reference Date*	1. Re-open 'Sample Timestamp' Variable Editor after saving the form. 2. Verify Lab Date setting.	Reference Date setting is retained for the Date and Time variable. No drift or reset occurs after form save or study re-open.	N/E
REG-RDV-06	Ref Date	P2-High	**Effective date comparison uses the Reference Date variable value correctly during data entry**	*Reference range 'WBC Q1 2025' has effective dates 01/01/2025–03/31/2025; 'Lab Collection Date' is the Reference Date variable*	1. Open a Lab form for subject S-200. 2. Select a lab. 3. Enter Lab Collection Date = 02/15/2025 (within effective range). 4. Enter WBC Count = 14.0 (above range). 5. Save.	Out-of-range flag triggers for WBC Count = 14.0 because 02/15/2025 falls within the 01/01/2025–03/31/2025 effective date window. The system correctly uses the Reference Date variable to determine range applicability.	N/E
REG-RDV-07	Ref Date	P1-Critical	**Effective date range is NOT applied when Lab Collection Date is outside the effective window**	*Reference range 'WBC Q1 2025' effective dates 01/01/2025–03/31/2025;*	1. Open a Lab form for subject S-201. 2. Enter Lab Collection Date = 06/15/2025. 3. Enter WBC Count = 14.0. 4. Save.	No out-of-range flag from 'WBC Q1 2025' appears. The range's effective dates do not include June 2025, so the range is not	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
				Lab Collection Date = 06/15/2025		applied. Other active ranges without effective dates may still trigger.	

REG-LAB Lab CRUD & State Transition Regression

Validates lab creation, detail editing, deactivation/reactivation state transitions, counter accuracy, and search behavior.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LAB-01	**Lab CRUD**	**P1-Critical**	**Creating multiple labs simultaneously does not create duplicate entries**	*No labs currently exist; user has Add/Edit Labs permission*	1. Navigate to Manage → Lab Management. 2. Enter three lab names on separate lines: 'Alpha Lab', 'Beta Lab', 'Gamma Lab'. 3. Click Create. 4. Check the labs list for duplicates.	Exactly three labs appear: Alpha Lab, Beta Lab, Gamma Lab. Labs (#) shows 3. No duplicates created. Each lab has its own independent DETAILS and LAB TESTS tabs.	**N/E**
REG-LAB-02	**Lab CRUD**	**P1-Critical**	**Lab code is unique per lab and does not persist to a newly created lab with the same name**	*Lab 'Alpha Lab' with code 'ALP-001' was deactivated; new 'Alpha Lab' is being created*	1. Create a new lab named 'Alpha Lab'. 2. Navigate to DETAILS tab for the new lab. 3. Check if any code is pre-populated.	New 'Alpha Lab' has an empty Lab Code field. It does not inherit the code 'ALP-001' from the deactivated lab with the same name. Codes must be explicitly entered.	**N/E**
REG-LAB-03	**Lab CRUD**	**P1-Critical**	**Lab details save does not affect reference ranges or subject data**	*Alpha Lab has 3 active reference ranges; 50 subjects have selected Alpha Lab in forms*	1. Navigate to Alpha Lab DETAILS tab. 2. Update the address to '999 New Address Ave, Chicago, IL 60601'. 3. Click Save. 4. Navigate to LAB TESTS → Manage Ranges. 5. Open subject S-050 Hematology Panel form.	Reference ranges remain unchanged — all 3 still active with correct limits. Subject S-050 data and out-of-range flags are unaffected. Only the address field in DETAILS reflects the new value.	**N/E**
REG-LAB-04	**Lab CRUD**	**P1-Critical**	**Labs (#) counter excludes deactivated labs and updates in real time**	*5 active labs, 2 deactivated; user deactivates one more active lab*	1. Note current Labs (#) = 5. 2. Deactivate 'Gamma Lab'. 3. Immediately check the Labs (#) label.	Labs (#) updates to 4 immediately after deactivation without page reload. Deactivated labs are never counted.	**N/E**
REG-LAB-05	**Lab CRUD**	**P2-High**	**Lab search returns correct results for partial name matches after lab rename**	*Lab was renamed from 'City Lab' to 'City Clinical Laboratory'*	1. Click the search icon in Lab Management. 2. Type 'City'. 3. Observe results.	Search returns 'City Clinical Laboratory' for the 'City' query. The old name is not returned. Partial match works on the updated name.	**N/E**
REG-LAB-06	**Lab CRUD**	**P2-High**	**Reactivated lab preserves all previously configured reference ranges**	*Beta Lab was deactivated after having 4 reference ranges; it is now being reactivated*	1. Navigate to Deactivated Labs → Reactivate Beta Lab. 2. Click Beta Lab → LAB TESTS → Manage Ranges.	All 4 reference ranges from before deactivation are present and active. No range configuration is lost during the	**N/E**

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						deactivation/reactivation lifecycle.	
REG-LAB-07	Lab CRUD	P1-Critical	**Out-of-Range flags in export remain linked to deactivated lab name**	*Subjects have OOR flags tied to 'Old Reference Lab' which was deactivated*	1. Export data via Analysis → Data Export with Include Labs. 2. Filter rows for subjects who used 'Old Reference Lab'. 3. Check the Lab Name column.	Lab Name column shows 'Old Reference Lab' even though the lab is now deactivated. Historical flag records are not anonymized or cleared. The lab name is preserved for audit and compliance.	N/E
REG-LAB-08	Lab CRUD	P2-High	**Export Labs includes both active and deactivated labs with their full range data**	*3 active labs, 2 deactivated; all have reference ranges*	1. Navigate to Lab Management → three-dot → Export Labs. 2. Open Excel file. 3. Count the lab tabs.	Excel file contains tabs for all 5 labs (active + deactivated). Deactivated labs are identifiable (status marked Inactive). Reference ranges for each lab are present in their respective tabs.	N/E
REG-LAB-09	Lab CRUD	P2-High	**Lab Details Audit Log captures lab reactivation event accurately**	*Beta Lab was deactivated and then reactivated*	1. Download Lab Details Audit Log. 2. Filter by Lab Name = 'Beta Lab'. 3. Find reactivation entry.	Audit log contains a row: Object = 'Lab Status', Action = 'Reactivated', Old Value = 'Inactive', New Value = 'Active', Changed By = (username), with accurate timestamp. Deactivation entry precedes it.	N/E
REG-LAB-10	Lab CRUD	P1-Critical	**Newly added lab (via + icon) is immediately available in the data entry Lab dropdown**	*A new lab 'Omega Diagnostics' is created mid-study after subjects already have data*	1. Add 'Omega Diagnostics' via the + icon in Lab Management. 2. Navigate to a subject's Hematology Panel form. 3. Click the Lab field dropdown.	'Omega Diagnostics' appears in the Lab dropdown immediately without any cache clearing or form refresh. Existing lab options are unaffected.	N/E

REG-RRC Reference Range — Data Integrity & Uniqueness Regression

Confirms range data persists correctly after form changes, lab edits, and deployment; uniqueness constraints remain enforced.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-RRC-01	Ref Ranges	P1-Critical	**Reference range lower/upper limits persist exactly after save — no rounding or truncation**	*Reference range for 'TSH' has lower=0.4, upper=4.0 (decimal values)*	1. Navigate to Lab Management → TSH reference range. 2. Note the lower and upper limits. 3. Navigate away and return to the same range.	Lower = 0.4, Upper = 4.0 persist exactly. No rounding to 0 or 4. No truncation of decimal values. Regression confirms numeric precision is maintained across saves and page loads.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-RRC-02	Ref Ranges	P1-Critical	**Multiple reference ranges for the same variable remain independent after one is edited**	*WBC Count has 3 ranges: 'WBC Adult Female', 'WBC Adult Male', 'WBC Pediatric'*	1. Edit 'WBC Adult Female' — change upper limit from 10.5 to 11.0. 2. Navigate to 'WBC Adult Male' and 'WBC Pediatric'. 3. Verify their limits.	'WBC Adult Male' and 'WBC Pediatric' retain their original limits unchanged. Editing one range does not cascade to sibling ranges for the same variable.	N/E
REG-RRC-03	Ref Ranges	P1-Critical	**Range name uniqueness is enforced at the variable level, not the lab level**	*Lab A has 'WBC High Range' for WBC Count; Lab B also exists*	1. Navigate to Lab B → LAB TESTS → WBC Count → Manage Ranges. 2. Attempt to create a range named 'WBC High Range'.	The name 'WBC High Range' is accepted in Lab B for WBC Count because uniqueness is enforced per variable within the same lab, not globally across all labs.	N/E
REG-RRC-04	Ref Ranges	P1-Critical	**Reference range survives when its parent lab's address and code are updated**	*Alpha Lab has 3 reference ranges; admin updates lab code and address*	1. Update Alpha Lab's lab code and address in DETAILS tab. Save. 2. Navigate to LAB TESTS → Manage Ranges. 3. Inspect each reference range.	All 3 reference ranges are unaffected by the lab detail update. Ranges remain active with correct limits, determinants, and effective dates.	N/E
REG-RRC-05	Ref Ranges	P1-Critical	**Deactivated reference range does not re-activate when its Lab form is unchecked and re-checked**	*WBC Count range 'WBC Pediatric' was deactivated; CBC Panel is being unchecked and re-selected*	1. In Lab Management → LAB TESTS, uncheck CBC Panel. 2. Save. 3. Re-check CBC Panel. 4. Click Manage Ranges → find 'WBC Pediatric'.	'WBC Pediatric' remains deactivated. Unchecking and re-checking the form does not reactivate deactivated ranges. Reactivation is permanent — once deactivated it cannot be reversed.	N/E
REG-RRC-06	Ref Ranges	P1-Critical	**Categorical Yes/No range value (Yes/No) does not change to numeric display after a system update**	*X-Ray Taken range 'X-Ray Expected' has expected value = Yes*	1. Navigate to the X-Ray Expected reference range after the latest system update. 2. Check the range value display.	Expected value still shows 'Yes' — not a number like 1 or true. Categorical Yes/No variable ranges are displayed as Yes/No in both Lab Management and Field Information popup.	N/E
REG-RRC-07	Ref Ranges	P2-High	**Reference range editor shows the correct Lab Test Display Label (not the system variable name)**	*Variable 'WBC_Measurement' has display label 'White Blood Cell Count (K/uL)'*	1. Navigate to Lab Management → Manage Ranges for the relevant form. 2. Check the label shown next to the range edit icon.	Manage Ranges displays 'White Blood Cell Count (K/uL)' as the label, not 'WBC_Measurement'. Display label is consistently used in Lab Management UI.	N/E
REG-RRC-08	Ref Ranges	P2-High	**Adding a second range via the + icon does not reset or clear the first range's limits**	*WBC Count has 'WBC Adult Range' with lower=4.5, upper=11.0; user adds a second range*	1. Navigate to WBC Count in Manage Ranges. 2. Click the + icon to add a new range. 3. Enter new range name and limits. 4. Save. 5. Revisit 'WBC Adult Range'.	'WBC Adult Range' retains lower=4.5, upper=11.0 after the second range is added. Adding a new range does not overwrite or reset existing ranges.	N/E
REG-RRC-09	Ref Ranges	P1-Critical	**Saving a range with lower limit > upper limit is blocked**	*Reference range editor for Hemoglobin is open*	1. Enter Name: 'Hgb Inverted Test'. 2. Set Lower Limit = 16.0, Upper Limit = 9.0 (lower > upper). 3. Click Save.	System blocks save with validation: 'Lower limit cannot exceed upper limit.' This prevents logically inverted ranges from	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						being applied during data entry flag evaluation.	
REG-RRC-10	Ref Ranges	P2-High	**Reference Range Audit Log records all three types of changes in one download**	*Within the same session: a range limit was updated, a determinant was added, effective dates were set*	1. Make the three changes to 'WBC Adult Range' in sequence. 2. Download Reference Range Audit Log.	All three changes appear as separate rows in the audit log: one for the limit update (Object = Upper Limit), one for determinant (Object = Range Determinant), one for effective date (Object = Start Date / End Date). Timestamps are sequential.	N/E
REG-RRC-11	Ref Ranges	P1-Critical	**Reference ranges are not affected by concurrent edits from two different users in the same lab**	*Two QA users (User A and User B) both have 'Manage Ranges' open for Alpha Lab simultaneously*	1. User A edits WBC Adult Range lower limit from 4.5 to 4.0 and saves. 2. User B (without refreshing) edits the same range upper limit from 11.0 to 12.0 and saves. 3. Reload the range and check both limit values.	Final state reflects both saves without data corruption. Either User B's save uses a conflict detection message or last-write-wins. The range is coherent — no mixed-up values from the concurrent edit.	N/E
REG-RRC-12	Ref Ranges	P2-High	**Range limits remain unaffected when LOINC code or unit is updated**	*TSH range has Lower=0.4, Upper=4.0; LOINC is being updated from '3016-3' to '11579-0'*	1. Open TSH reference range → update LOINC to '11579-0'. 2. Save. 3. Revisit the range limits.	Lower=0.4 and Upper=4.0 are unchanged after updating the LOINC code. Updating metadata (LOINC, unit) does not affect numeric range values.	N/E

REG-DET Range Determinant Logic — Cross-Form & Visit Regression

🔍 *Validates determinant evaluation correctness for cross-form, multi-visit, and ePRO exclusion scenarios.*

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-DET-01	Determinant	P1-Critical	**Determinant value updated mid-study retrospectively re-evaluates the correct range for new entries**	*Subject S-300 had Gender=Male (WBC Male Range applied); Gender is corrected to Female*	1. Update Gender for S-300 from Male to Female. 2. Open the Hematology Panel Lab form for S-300. 3. Enter WBC Count = 11.0 (above Female upper limit 10.5, but within Male limit 11.0). 4. Save.	Out-of-range flag triggers for WBC = 11.0 using the Female range (10.5 upper limit). The corrected Gender=Female is applied to new entries. Historical flags before the correction remain unchanged.	N/E
REG-DET-02	Determinant	P1-Critical	**Determinant from a different form visit is applied correctly to the Lab form flag evaluation**	*Age (Range Determinant) is configured to be checked at Screening visit; lab data is entered at Week 12*	1. Enter Age=25 at Screening visit for subject S-301. 2. At Week 12, open CBC Panel → select lab → enter WBC=14.0. 3. Save.	Flag evaluation uses the Screening visit Age=25 value (as configured). The appropriate adult range fires for Week 12 data. Visit-specific determinant lookup is correct.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-DET-03	Determinant	P1-Critical	**Determinant entered via ePRO form does NOT trigger Lab Management flag evaluation**	*Gender entered via ePRO Patient Diary for S-302; same Gender variable is the Range Determinant*	1. Subject S-302 has Gender=Female entered ONLY via the ePRO form. 2. Open the Hematology Panel in EDC for S-302. 3. Enter WBC=12.0 (above Female upper limit 10.5). 4. Save.	No out-of-range flag appears. ePRO-sourced determinant values are excluded from Lab Management evaluation. Only values entered in EDC trigger flag logic — this is a critical regression check post-patch.	N/E
REG-DET-04	Determinant	P1-Critical	**Changing a determinant variable value in EDC causes next data entry to use the updated range**	*S-303 had Pregnancy Status=No (non-pregnant range applied); status updated to Yes in EDC*	1. Update Pregnancy Status for S-303 from No to Yes in EDC. 2. Open Lab form → select lab → enter a Lab Test Result value. 3. Save.	The next data entry evaluates using the Pregnancy=Yes range (if configured). Flag state reflects the updated determinant. The previous entry's flag state remains as a historical record.	N/E
REG-DET-05	Determinant	P1-Critical	**'At any time' determinant option fires when the determinant value exists at ANY visit**	*Pregnancy Status uses 'At any time' option; S-304 has Pregnancy=Yes at Week 4 only*	1. Enter Lab Test Result value at the Screening visit for S-304 (before Week 4). 2. Later enter Lab Test Result at Week 8 visit. 3. Check whether flag evaluates using the Pregnancy=Yes range at both visits.	At Week 8: Flag uses Pregnancy=Yes range because the determinant value exists somewhere in the study ('at any time'). Behavior is consistent regardless of which visit the determinant was entered.	N/E
REG-DET-06	Determinant	P2-High	**Determinant variable deletion deactivates its associated reference ranges**	*Gender Range Determinant is used in 3 reference ranges; variable is about to be deleted*	1. Delete the 'Gender' variable from its form. 2. Navigate to Lab Management → Manage Ranges → check ranges that used Gender as determinant.	All 3 ranges that used Gender as their determinant condition are deactivated. The reference range section shows them as inactive. Non-Gender ranges remain active.	N/E
REG-DET-07	Determinant	P2-High	**Multiple determinants on the same reference range all must match for the flag to trigger**	*Range 'WBC Pregnant Female' requires Gender=Female AND Pregnancy=Yes*	1. S-305 has Gender=Female but Pregnancy=No. 2. Enter WBC=13.0. 3. Save. 4. S-306 has Gender=Female AND Pregnancy=Yes. 5. Enter WBC=13.0 for S-306.	S-305: No flag — both conditions (Gender AND Pregnancy) must match. S-306: Flag triggers — both Gender=Female and Pregnancy=Yes are met. Multi-determinant AND logic is correctly enforced.	N/E
REG-DET-08	Determinant	P1-Critical	**Determinant check uses the visit-specific snapshot, not the most recent visit's value**	*Age is configured as Range Determinant at Screening; S-307 entered Age=17 at Screening, but is now 19*	1. S-307 has Age=17 entered at Screening. 2. Enter a lab value at a follow-up visit. 3. Check which age-based range fires.	The system evaluates using Age=17 from the Screening visit (as configured), not the patient's current age. Pediatric range fires even though the patient is now 19. Visit-specific determinant snapshot is respected.	N/E
REG-DET-09	Determinant	P2-High	**Units Range Determinant correctly limits range**	*TSH has two ranges: one for mIU/L (Lower=0.4, Upper=4.0) and one for*	1. S-308: Units=mIU/L, TSH=5.0 (above 4.0 in mIU/L range). Enter and save. 2.	S-308: Out-of-range flag fires using mIU/L range (5.0 > 4.0). S-309: Flag fires using uIU/mL	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
			evaluation to the specified unit type	*uIU/mL (Lower=0.4, Upper=4.0 equivalent); Units variable is Range Determinant*	S-309: Units=uIU/mL, TSH=5.0. Enter and save.	range. Unit-specific ranges evaluate independently. No cross-contamination of range evaluation between unit types.	

REG-EFD Effective Date — Validation & Bulk Override Regression

Validates effective date persistence, bulk overwrite warnings, and cross-lab isolation hold after system changes.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-EFD-01	Eff Dates	P2-High	**Effective dates persist after the parent reference range's limits are updated**	*WBC Adult Range has effective dates 01/01/2025–12/31/2025; limits are being updated*	1. Open 'WBC Adult Range' → change upper limit from 11.0 to 11.5 → save. 2. Revisit the range → check Effective Dates section.	Effective dates 01/01/2025–12/31/2025 remain unchanged after the limit update. Updating range limits does not reset or clear effective dates.	N/E
REG-EFD-02	Eff Dates	P2-High	**Bulk effective date apply generates warning when at least one selected range already has dates**	*Alpha Lab: 'WBC Adult Range' has dates 01/01/2025–12/31/2025; 'RBC Adult Range' has no dates*	1. Open Effective Dates panel for Alpha Lab. 2. Select both 'WBC Adult Range' and 'RBC Adult Range'. 3. Enter new dates 06/01/2025–12/31/2025. 4. Click Apply.	Warning appears: 'One or more selected ranges already have effective dates. Applying will overwrite them.' After confirmation: both ranges get 06/01/2025–12/31/2025. The warning is shown before overwrite — not after.	N/E
REG-EFD-03	Eff Dates	P1-Critical	**Calendar icon appears next to lab test only when at least one of its ranges has effective dates**	*WBC Count has two ranges: 'WBC Adult Range' with effective dates and 'WBC Pediatric Range' without*	1. Deactivate 'WBC Adult Range' (the one with effective dates). 2. Check the LAB TESTS tab for the calendar icon next to WBC Count.	Calendar icon disappears from WBC Count in the LAB TESTS tab after the only dated range is deactivated. When 'WBC Pediatric Range' (no dates) is the only active range, no calendar icon shows.	N/E
REG-EFD-04	Eff Dates	P2-High	**Effective dates from Lab A cannot be bulk-applied to Lab B via the same Apply action**	*Alpha Lab and Beta Lab both exist; user is in Alpha Lab's Effective Dates panel*	1. Open Alpha Lab → LAB TESTS → Effective Dates panel. 2. Select multiple Alpha Lab ranges. 3. Look for any option to cross-apply to Beta Lab's ranges in the same panel.	No option exists to apply to Beta Lab ranges from Alpha Lab's Effective Dates panel. The panel is lab-scoped. User must navigate to Beta Lab separately to configure its ranges.	N/E
REG-EFD-05	Eff Dates	P2-High	**Effective date audit entries are captured per bulk apply action correctly**	*Bulk effective dates applied to 3 ranges in Alpha Lab simultaneously*	1. Apply dates 03/01/2025–09/30/2025 to 3 ranges via bulk panel. 2. Download Reference Range Audit Log.	Audit log has 6 rows (2 per range — Start Date and End Date) all with the same timestamp (or within seconds). Each row	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						references the correct range name, lab, and changed-by username.	
REG-EFD-06	**Eff Dates**	**P2-High**	**Effective date validation mm/dd/yyyy is enforced even when the browser locale is non-US**	*Browser locale set to UK (dd/mm/yyyy); effective date editor is open*	1. Set browser locale to UK. 2. Open an effective date field. 3. Enter a date in dd/mm/yyyy format: '31/01/2025'. 4. Click Save.	System enforces mm/dd/yyyy regardless of browser locale. '31/01/2025' is either rejected (month 31 is invalid) or interpreted incorrectly. System should reject and prompt for mm/dd/yyyy format.	**N/E**
REG-EFD-07	**Eff Dates**	**P2-High**	**Ranges without effective dates remain in scope for flag evaluation regardless of Lab Collection Date**	*WBC Adult Range has NO effective dates; Lab Collection Date is entered for the subject*	1. Enter Lab Collection Date = 05/20/2025 for subject S-400. 2. Enter WBC = 14.0. 3. Save.	WBC Adult Range evaluates for the flag because it has no effective date restriction. Ranges without effective dates are always in scope. Flag triggers for WBC=14.0.	**N/E**
REG-EFD-08	**Eff Dates**	**P1-Critical**	**Simultaneous overlapping effective dates from two ranges both evaluate but only first displays inline**	*WBC Count: Range A effective 01/01/2025–06/30/2025 and Range B effective 04/01/2025–12/31/2025 — both active on 05/15/2025*	1. Enter Lab Collection Date = 05/15/2025. 2. Enter WBC = 2.0 (below both ranges' lower limits). 3. Save. 4. Count the flag icons and open Field Information popup.	Multiple flag icons appear. Field Information popup lists both Range A and Range B as triggered. Inline display below the variable shows only the first applicable range. Behavior is consistent with non-date overlapping ranges.	**N/E**
REG-EFD-09	**Eff Dates**	**P2-High**	**Clearing both effective date fields (Start and End) from an existing range removes the Calendar icon**	*WBC Adult Range has effective dates 01/01/2025–12/31/2025*	1. Open WBC Adult Range → clear Start Date and End Date fields. 2. Click Save. 3. Check LAB TESTS tab for Calendar icon next to WBC Count.	Calendar icon disappears from WBC Count after clearing the effective dates. Range still applies for flag evaluation (no date restriction). Removing dates does not deactivate the range.	**N/E**
REG-EFD-10	**Eff Dates**	**P2-High**	**Search in bulk Effective Dates panel returns correct ranges after a range rename**	*Reference range 'WBC Standard' was renamed to 'WBC Adult Standard 2025'*	1. Open Effective Dates bulk panel for the lab. 2. Type 'WBC Standard' in the search field. 3. Then type 'WBC Adult Standard 2025'.	Search for 'WBC Standard' returns no results (old name). Search for 'WBC Adult Standard 2025' returns the renamed range. Search is live against current range names.	**N/E**

REG-LCU LOINC & Unit Data Persistence Regression

Verifies LOINC codes and units persist across saves, appear in all export channels, and are audited correctly.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LCU-01	LOINC	P2-High	**LOINC code persists after the reference range limits are updated**	*WBC Count range has LOINC '26464-8'; lower limit is being changed*	1. Update lower limit from 4.5 to 4.0. 2. Save. 3. Revisit the reference range and check LOINC field.	LOINC code '26464-8' persists after the limit update. Updating numeric range values does not clear the LOINC metadata.	N/E
REG-LCU-02	LOINC	P2-High	**LOINC code change is captured accurately in the Reference Range Audit Log**	*WBC Count LOINC changed from '26464-8' to '6690-2' by user qa_manager*	1. Change LOINC for WBC Count range. 2. Download Reference Range Audit Log. 3. Locate the LOINC update row.	Audit row: Object = LOINC, Action = Updated, Old Value = 26464-8, New Value = 6690-2, Changed By = qa_manager, with correct timestamp. Both old and new values are captured.	N/E
REG-LCU-03	LOINC	P2-High	**Unit value appears in Data Export (Include Labs) for all subjects with lab data in that range**	*WBC Count has unit 'K/uL'; 30 subjects have WBC data in Alpha Lab*	1. Export data via Analysis → Data Export → select Hematology Panel → Include Labs → Download. 2. Open Excel → find WBC Count unit column.	All 30 subject rows show 'K/uL' in the unit column for WBC Count. No blank or mismatched unit values for subjects in Alpha Lab.	N/E
REG-LCU-04	LOINC	P2-High	**Entering unit only (no LOINC) is valid and does not create a validation error**	*Reference range for 'Hemoglobin' has no LOINC code; unit 'g/dL' is being added*	1. Enter Lab Test Unit = 'g/dL' for Hemoglobin range. 2. Leave LOINC field blank. 3. Save.	Range saves successfully with unit='g/dL' and no LOINC. No validation error. The unit appears in Export Labs and Data Export. LOINC field remains blank in exports.	N/E
REG-LCU-05	LOINC	P2-High	**Deactivated range's LOINC and unit still appear in Export Labs for audit purposes**	*TSH range 'TSH Standard' was deactivated; it had LOINC '3016-3' and unit 'mIU/L'*	1. Export Labs → open Excel → navigate to TSH Standard range row.	Deactivated range row includes LOINC = '3016-3' and Unit = 'mIU/L'. Historical LOINC and unit metadata is preserved in exports for deactivated ranges.	N/E
REG-LCU-06	LOINC	P2-High	**LOINC code with non-standard characters is accepted or rejected consistently**	*User attempts to enter LOINC with special characters: '26464-8!' or '26464 8'*	1. Open WBC Count reference range. 2. Enter LOINC = '26464-8!' (with exclamation). 3. Save. 4. Enter LOINC = '26464 8' (with space). 5. Save.	System either accepts or rejects the entries consistently based on LOINC format rules (alphanumeric and hyphen only). Special characters like '!' are rejected. Behavior is consistent — not accepted in one session and rejected in another.	N/E

REG-CDR Copy & Deactivate Reference Range Regression

Validates copy fidelity, deactivation permanence, and historical flag preservation after range deactivation.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-CDR-01	Copy/Deact	P2-High	**Copied reference range is fully independent — editing the copy does not affect the original**	*'CD4 Female Low' was copied to 'CD4 Female Low (Copy)' in Alpha Lab*	1. Open 'CD4 Female Low (Copy)' → change lower limit from 100 to 150. 2. Save. 3. Open original 'CD4 Female Low' → verify lower limit.	Original 'CD4 Female Low' retains lower = 100. The copy is an independent entity — no shared state with the source after creation.	N/E
REG-CDR-02	Copy/Deact	P2-High	**Copied range name defaults to a unique name and cannot be manually renamed to the source name**	*'WBC Adult Normal' was copied*	1. Open the copied range. 2. Attempt to rename it to 'WBC Adult Normal' (matching source). 3. Save.	System blocks the rename: 'Range name must be unique for this variable.' Copied range must have a distinct name within the same lab and variable scope.	N/E
REG-CDR-03	Copy/Deact	P2-High	**Copied range effective dates are independent — changing source dates does not affect copy**	*'CD4 Female Low' effective dates 01/01/2025–12/31/2025 were copied to the copy range*	1. Update 'CD4 Female Low' effective dates to 06/01/2025–12/31/2025. 2. Open 'CD4 Female Low (Copy)' → check its effective dates.	Copied range retains original dates 01/01/2025–12/31/2025. Updating source range dates does not propagate to the copy.	N/E
REG-CDR-04	Copy/Deact	P1-Critical	**Deactivated reference range flags on historical subject data remain in the export**	*10 subjects had OOR flags under 'WBC Pediatric Range'; range is now deactivated*	1. Export data with Include Labs → filter subjects who triggered 'WBC Pediatric Range'. 2. Check Out-of-Range and Flag State columns.	All 10 subjects' historical flag data is present in the export. Deactivating the range does not purge historical flag records. Flag State column shows 'Above' or 'Below' for affected rows.	N/E
REG-CDR-05	Copy/Deact	P1-Critical	**Deactivated range does not trigger for new data entry even if the value is outside the former range limits**	*'WBC Pediatric Range' is deactivated; a new subject S-500 enters WBC=2.0*	1. Open Hematology Panel for S-500. 2. Select a lab that had 'WBC Pediatric Range'. 3. Enter WBC=2.0. 4. Save.	No flag appears for WBC=2.0. 'WBC Pediatric Range' is deactivated and not evaluated. Only active ranges trigger flags. Regression confirms deactivation is enforced for all new data.	N/E
REG-CDR-06	Copy/Deact	P1-Critical	**Reactivation of reference range is truly impossible — no UI or API path exists**	*'WBC Pediatric Range' was deactivated*	1. Navigate to the deactivated range in Lab Management. 2. Inspect three-dot menu for 'Reactivate' option. 3. Attempt via direct URL or API call to reactivate the range.	No 'Reactivate' option exists in the UI three-dot menu for deactivated reference ranges. API-level attempts return an error. Deactivation is permanent and irreversible by design.	N/E
REG-CDR-07	Copy/Deact	P2-High	**Copying a range with LOINC and unit preserves both in the copy**	*'WBC Adult Normal' has LOINC='26464-8' and unit='K/uL'*	1. Copy 'WBC Adult Normal'. 2. Open the copied range. 3. Check LOINC and unit fields.	Copied range shows LOINC='26464-8' and unit='K/uL'. All metadata is carried over to the copy, identical to the source at the time of copying.	N/E
REG-CDR-08	Copy/Deact	P2-High	**Range deactivation is recorded in the Reference Range Audit Log with full context**	*'Platelet Adult Range' was deactivated by user qa_lead*	1. Download Reference Range Audit Log. 2. Search for 'Platelet Adult Range' deactivation entry.	Audit row: Lab Name = Alpha Lab, Range Name = Platelet Adult Range, Object = Range Status, Action = Deactivated, Old Value =	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						Active, New Value = Inactive, Changed By = qa_lead, with correct timestamp.	

REG-LDE Data Entry — Lab Selection, Change & Persistence

Verifies lab field mandatory enforcement, lab change persistence, range display accuracy, and cross-visit independence.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-LDE-01	Data Entry	P1-Critical	**Lab field retains its selection after partial form save failure (e.g., required field missing)**	*Subject S-600 opens Hematology Panel; selects 'Alpha Lab'; WBC field is left blank (required)*	1. Select Alpha Lab in the Lab field. 2. Leave WBC Count blank. 3. Attempt Save → observe validation error. 4. Fill WBC Count = 7.0 → Save again.	After the first failed save, Alpha Lab remains selected in the Lab field. User does not need to re-select the lab after a failed save attempt. Second save completes successfully with Alpha Lab retained.	N/E
REG-LDE-02	Data Entry	P1-Critical	**Changing lab mid-study updates the inline reference range display immediately after Change Lab**	*S-601 has Alpha Lab saved on Hematology Panel (WBC range 4.5–11.0); switching to Beta Lab (WBC range 4.0–10.8)*	1. Open S-601's Hematology Panel. 2. Change Lab field to Beta Lab. 3. Click 'Change Lab' button. 4. Observe the reference range displayed below WBC Count.	After Change Lab: reference range below WBC Count updates from '4.5–11.0' to '4.0–10.8' immediately. Old range is no longer displayed. The form reflects Beta Lab's ranges for all Lab Test Result variables.	N/E
REG-LDE-03	Data Entry	P1-Critical	**Previous lab's out-of-range flag is cleared when lab is changed and new value is within new range**	*S-602 had WBC=12.0 flagged under Alpha Lab (range 4.5–11.0); lab changed to Beta Lab (range 4.0–12.5)*	1. Open S-602's Hematology Panel. 2. Change lab to Beta Lab → click Change Lab. 3. Confirm WBC=12.0 is still in the form. 4. Verify flag status.	Flag is no longer active because WBC=12.0 is within Beta Lab's range (4.0–12.5). Flag transitions to Inactive. Field Information popup shows status change. The alpha lab flag record is archived in the audit, not deleted.	N/E
REG-LDE-04	Data Entry	P1-Critical	**Lab selection search is case-insensitive**	*Multiple labs: 'Alpha Lab', 'beta lab' (lowercase in system), 'GAMMA LAB' (uppercase)*	1. Open a Lab form → click Lab field. 2. Type 'alpha'. 3. Type 'BETA'. 4. Type 'Gamma'.	Search returns the matching lab regardless of case entered. 'alpha', 'ALPHA', 'Alpha' all return 'Alpha Lab'. Case-insensitive search is enforced consistently across environments.	N/E
REG-LDE-05	Data Entry	P1-Critical	**Data entry is blocked when no active lab exists in the study**	*All labs have been deactivated; no active lab is available*	1. Navigate to a subject's Hematology Panel Lab form. 2. Click the Lab field. 3. Inspect available options.	Lab field dropdown is empty. A message indicates no labs are available. Data cannot be saved. System does not allow selecting a	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						blank lab or bypassing the lab requirement.	
REG-LDE-06	Data Entry	P2-High	**Reference range displays correctly for each independent visit (Screening and Week 4 have different labs selected)**	*S-603: Screening visit has Alpha Lab (WBC 4.5–11.0); Week 4 visit has Beta Lab (WBC 4.0–10.8)*	1. Open S-603's Screening Hematology Panel → check WBC range displayed. 2. Open S-603's Week 4 Hematology Panel → check WBC range displayed.	Screening: WBC range shows '4.5–11.0' (Alpha Lab). Week 4: WBC range shows '4.0–10.8' (Beta Lab). Each visit's form independently uses the lab selected for that visit.	N/E
REG-LDE-07	Data Entry	P1-Critical	**Saving a Lab form with a value exactly at the boundary (upper=11.0, entered=11.0) shows no flag**	*Alpha Lab WBC range 4.5–11.0; subject S-604 enters WBC=11.0*	1. Select Alpha Lab for S-604. 2. Enter WBC Count = 11.0. 3. Save.	No out-of-range flag appears for WBC=11.0. The upper boundary is inclusive. Reference range '4.5–11.0' is displayed below the field without a flag icon.	N/E
REG-LDE-08	Data Entry	P1-Critical	**Flag triggers for value 0.001 above the upper boundary — minimal out-of-range is flagged**	*Alpha Lab WBC range lower=4.5, upper=11.0; subject S-605 enters WBC=11.1*	1. Enter WBC = 11.1 for S-605 → Save.	Higher out-of-range flag appears. Even a minimal exceedance of 0.1 above 11.0 triggers the flag. The system does not have a tolerance/grace value — any value strictly above the upper limit is flagged.	N/E
REG-LDE-09	Data Entry	P2-High	**Updating a Lab form field other than the Lab Test Result variable does not re-trigger flags**	*S-606 has WBC=7.0 (within range, no flag); form has a text variable 'Sample Note'*	1. Open S-606's Hematology Panel. 2. Update 'Sample Note' from 'AM sample' to 'PM sample'. 3. Save.	No out-of-range flag is added for WBC Count (already within range). The save of a non-lab-test-result variable does not re-evaluate or create new flags. Existing Inactive flags remain Inactive.	N/E
REG-LDE-10	Data Entry	P2-High	**Bulk-uploaded lab data respects the same mandatory lab field requirement as manual entry**	*Bulk upload file is missing a lab name for 3 out of 20 rows*	1. Upload the bulk file. 2. Review upload results report.	The 3 rows without a lab name fail with per-row error: 'Lab field is required for Lab forms.' The other 17 rows upload successfully. Error log identifies the specific subject rows with missing lab.	N/E
REG-LDE-11	Data Entry	P1-Critical	**Clearing and re-entering a Lab Test Result value within range removes the active flag**	*S-607 has WBC=14.0 with active higher flag; value cleared and re-entered as 7.0*	1. Open S-607's Hematology Panel. 2. Clear WBC Count field. 3. Enter WBC = 7.0. 4. Save.	WBC=7.0 is within range (4.5–11.0). Higher flag becomes Inactive. Flag icon disappears. Field Information popup shows Status = Inactive with original activation and resolution timestamps.	N/E

REG-OOR Out-of-Range Flag — Trigger Accuracy & State Regression

Validates flag precision, multi-range conflict resolution, deactivated range exclusion, and flag state export accuracy.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-OOR-01	OOR Flags	P1-Critical	**Value exactly 0.001 below lower limit triggers lower out-of-range flag**	*Alpha Lab WBC range: Lower=4.5, Upper=11.0; subject S-700 enters WBC=4.4*	1. Enter WBC=4.4 for S-700 → Save.	Lower out-of-range flag appears. WBC=4.4 is below the lower limit 4.5. Flag icon (downward indicator) is visible. Reference range '4.5–11.0' displayed below the field.	N/E
REG-OOR-02	OOR Flags	P1-Critical	**Updating range limits for an active flag re-evaluates whether the flag should remain Active or become Inactive**	*S-701 has WBC=11.5 (above 11.0 upper limit — flag Active); upper limit is now increased to 12.0*	1. Update Alpha Lab WBC upper limit from 11.0 to 12.0. 2. Navigate to S-701's Hematology Panel and refresh or re-save the form.	After the range limit update, WBC=11.5 is now within range (≤12.0). Flag for S-701 transitions to Inactive. System re-evaluates existing data against updated limits upon next form interaction.	N/E
REG-OOR-03	OOR Flags	P1-Critical	**When multiple reference ranges fire, the first range in the list is shown inline and all appear in the popup**	*WBC Count has Range A (3.0–7.0, listed first) and Range B (5.0–10.0, listed second); value = 2.0*	1. Enter WBC=2.0 → Save. 2. Check inline range display below the field. 3. Click flag icon → view Field Information popup.	Inline display shows Range A's limits (3.0–7.0) only. Field Information popup lists both Range A and Range B as triggered. Multiple flag icons visible. First applicable range = first in list order.	N/E
REG-OOR-04	OOR Flags	P1-Critical	**Categorical Yes/No flag shows 'Not Equal' status in export when expected=Yes but No is entered**	*X-Ray Taken range: Expected=Yes; subject S-702 entered No*	1. Export data with Include Labs for Imaging Panel → open Excel → find S-702's X-Ray Taken row.	Out-of-Range column shows 'Not Equal' for S-702. Flag State column reflects 'Not Equal'. This is distinct from numeric 'Above'/'Below' — categorical ranges use Equal/Not Equal.	N/E
REG-OOR-05	OOR Flags	P1-Critical	**Flag for deactivated lab does not reactivate when a new range is created under the same lab name after reactivation**	*S-703 had WBC=14.0 flagged under 'Beta Lab'; Beta Lab was deactivated and reactivated; new WBC range created*	1. Deactivate Beta Lab (existing flag on S-703 becomes historical). 2. Reactivate Beta Lab. 3. Create a new WBC range (3.0–15.0) in Beta Lab. 4. Open S-703's form → check WBC=14.0 flag status.	S-703's historical flag for WBC=14.0 under the OLD range remains Inactive (range was deactivated). The new range (3.0–15.0) evaluates WBC=14.0 as within range — no new flag. Old and new ranges are treated independently.	N/E
REG-OOR-06	OOR Flags	P1-Critical	**Out-of-Range flag state in export matches the live form flag icon state for the same subject**	*S-704 has WBC=13.0 (Active flag); S-705 has WBC=7.0 (within range, no flag); S-706 had WBC=15.0 corrected to 6.0 (Inactive flag)*	1. Export with Include Labs for Hematology Panel. 2. Compare Out-of-Range column for S-704, S-705, S-706 against live forms.	S-704: Export = 'Above', form = Active higher flag. S-705: Export = 'Within', form = no flag. S-706: Export = 'Within', form = no flag icon (Inactive). Export and live form are in perfect sync.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-OOR-07	OOR Flags	P2-High	**Adding a new reference range to an existing lab test variable does not reset flags for existing ranges**	*S-707 has an active flag from 'WBC Adult Range'; a new range 'WBC Elderly Range' is added to WBC Count*	1. Add 'WBC Elderly Range' to WBC Count. 2. Navigate to S-707's Hematology Panel → check existing WBC Adult Range flag.	Active flag from 'WBC Adult Range' remains Active on S-707's form. Adding a new range does not disturb existing flag states for previously configured ranges.	N/E
REG-OOR-08	OOR Flags	P1-Critical	**Flag does not trigger when Lab Test Result variable field is saved empty**	*Alpha Lab WBC range configured; subject S-708 saves Hematology Panel with WBC Count blank*	1. Select Alpha Lab. 2. Leave WBC Count blank (if field is optional). 3. Save.	No out-of-range flag appears for an empty WBC Count field. Flag evaluation requires a value to be present. Blank/null values are excluded from range comparison.	N/E
REG-OOR-09	OOR Flags	P2-High	**Out-of-Range flag icon is visually distinct for Above vs Below ranges**	*Two subjects: S-709 (WBC=14.0, above) and S-710 (WBC=2.0, below)*	1. View Hematology Panel for S-709 → note the flag icon type. 2. View Hematology Panel for S-710 → note the flag icon type.	S-709 shows the higher (upward) flag icon. S-710 shows the lower (downward) flag icon. The two icons are visually distinct — a QA engineer can differentiate Above from Below without opening the popup.	N/E
REG-OOR-10	OOR Flags	P1-Critical	**Correcting one variable's out-of-range value does not affect flags for other variables on the same form**	*S-711: WBC=14.0 (Active flag) and Hemoglobin=18.0 (Active flag); WBC is corrected to 7.0*	1. Update WBC from 14.0 to 7.0 → Save. 2. Check Hemoglobin flag.	WBC flag becomes Inactive. Hemoglobin flag remains Active (18.0 is still out of range). Correcting one variable's flag does not cascade to other variables on the same form.	N/E
REG-OOR-11	OOR Flags	P1-Critical	**Flag evaluation respects updated reference range limits applied after initial data entry**	*WBC Adult Range was 4.5–11.0; range updated to 4.5–12.0; subject S-712 had WBC=11.5 (was Active flag)*	1. Update WBC Adult Range upper limit to 12.0. 2. Open S-712's form → refresh or re-save.	After limit update, WBC=11.5 is within 4.5–12.0. S-712's flag transitions from Active to Inactive. System re-evaluates historical data against the new limits upon form interaction.	N/E
REG-OOR-12	OOR Flags	P2-High	**Flag triggered by a determinant-matched range and a non-determinant range simultaneously shows both in popup**	*WBC Count: Range A (no determinant, 4.5–11.0) and Range B (Gender=Female determinant, 4.0–10.5); S-713 is Female, WBC=11.2*	1. S-713 has Gender=Female. 2. Enter WBC=11.2 → Save. 3. Open Field Information popup.	Range A fires (11.2 > 11.0) and Range B fires (11.2 > 10.5 for Female). Multiple flags appear. Popup shows both Range A and Range B as triggered with their respective expected ranges and labs.	N/E

REG-FIP Field Info Popup — Data Accuracy & Status Transition

Validates popup field accuracy, Active-to-Inactive state transitions, and multi-range popup listing.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-FIP-01	FI Popup	P2-High	**Field Information popup timestamp matches the form save time within 60 seconds**	*Subject S-800 triggers an OOR flag at a known time (e.g., 14:32 local)*	1. Note the system clock at save time. 2. Click the flag icon. 3. Compare the popup timestamp to the noted save time.	Popup timestamp matches the save time within 60 seconds. Timezone shown matches the study-configured timezone (not UTC unless study is UTC). Timestamp precision is accurate.	N/E
REG-FIP-02	FI Popup	P2-High	**Popup shows Inactive status immediately after corrective value save — no page reload needed**	*S-801 has Active flag (WBC=14.0); value corrected to 7.0 in the same session*	1. Open Hematology Panel for S-801. 2. Update WBC to 7.0 → Save. 3. Without reloading the page, click the former flag area.	Popup opens and shows Status = Inactive immediately after save, in the same browser session. No full page reload is required for the status to reflect the correction.	N/E
REG-FIP-03	FI Popup	P1-Critical	**Popup lists all ranges that fired for overlapping effective date windows**	*Two ranges with overlapping effective dates both fire for S-802 on 05/15/2025*	1. Open Hematology Panel for S-802 → click flag icon for WBC Count.	Popup lists both ranges — each with its own name, expected limits, lab name, and Active status. Both overlap-triggered ranges appear as separate entries in the popup.	N/E
REG-FIP-04	FI Popup	P2-High	**Popup for Categorical Yes/No range shows Expected Value as Yes or No — not numeric**	*X-Ray Taken Expected=Yes; S-803 entered No → Active flag*	1. Click the out-of-range flag icon next to X-Ray Taken for S-803.	Popup shows: Expected Range/Value = 'Yes', Entered = 'No', Status = Active. No numeric representation (e.g., 1 or 0) is shown. Categorical values are displayed as their labels.	N/E
REG-FIP-05	FI Popup	P2-High	**Popup displays the correct lab name even after the lab's display name is updated**	*S-804 triggered a flag under 'Alpha Lab'; lab display name was later changed to 'Alpha Clinical Lab'*	1. Click the flag icon for S-804's WBC Count.	Popup shows the lab name at the time of flag creation ('Alpha Lab') OR the current lab name ('Alpha Clinical Lab'), consistently. Regression confirms popup lab name does not show a blank or incorrect lab.	N/E
REG-FIP-06	FI Popup	P2-High	**Field Information popup is read-only — no editable fields or action buttons inside**	*Any subject with an active OOR flag; popup is open*	1. Open the Field Information popup for an active flag. 2. Attempt to edit any field inside the popup. 3. Look for action buttons (e.g., Acknowledge, Override).	Popup is entirely read-only. No editable fields, no action buttons. It is an information display only. No data can be submitted from within the popup.	N/E
REG-FIP-07	FI Popup	P1-Critical	**Popup shows correct range name after range was renamed post-flag creation**	*WBC Adult Range was renamed to 'WBC Adult Standard 2025' AFTER S-805 triggered a flag*	1. Click flag icon on S-805's form. 2. Note the range name in the popup.	Popup shows either the original name ('WBC Adult Range') or the current name ('WBC Adult Standard 2025') — consistently. It does not show a blank, broken reference, or mismatched name.	N/E

REG-UPL Upload — Mapping Persistence & Flag Regression

Validates bulk upload field mapping is durable, lab column mapping is correctly applied, and flags fire post-upload.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-UPL-01	Upload	P2-High	**Saved bulk upload mapping retains the Lab field mapping after a session expiry**	*A field mapping for CBC Panel was created and saved in a previous session*	1. Log out and log back in. 2. Navigate to Configure → Bulk Upload → find the CBC Panel mapping. 3. Open the mapping and verify the Lab field assignment.	Lab field mapping persists across sessions. The column from the upload file is still correctly mapped to the Lab field. No re-mapping is required after session expiry.	N/E
REG-UPL-02	Upload	P1-Critical	**Bulk upload with an invalid lab name in one row fails that row while others succeed**	*Upload CSV has 19 valid rows with 'Alpha Lab' and 1 row with 'UnknownLab'*	1. Upload the 20-row file via Bulk Upload. 2. Review the upload results report.	19 rows upload successfully. The UnknownLab row fails with error: 'Lab name not found in Lab Management.' Error log identifies the row number. Partial upload succeeds — invalid rows don't block valid rows.	N/E
REG-UPL-03	Upload	P1-Critical	**Out-of-range flags triggered by bulk upload match flags from manual entry for the same values**	*Manual entry: WBC=14.0 in Alpha Lab → Active flag. Bulk upload: same subject, same WBC=14.0 via bulk for a different visit.*	1. Upload file with WBC=14.0 for S-900 at Week 4. 2. Open CBC Panel for S-900 at Week 4.	Flag behavior is identical to manual entry: higher OOR flag appears for WBC=14.0 in Alpha Lab. Bulk upload does not bypass range evaluation. Flags are applied consistently.	N/E
REG-UPL-04	Upload	P2-High	**Re-uploading corrected data for a subject updates the field and transitions the flag to Inactive**	*S-901 had WBC=14.0 uploaded (Active flag); corrected file re-uploaded with WBC=7.0*	1. Re-upload with corrected WBC=7.0 for S-901. 2. Open CBC Panel for S-901.	WBC Count field shows 7.0. Higher OOR flag is now Inactive. Field Information popup shows Status = Inactive. The data overwrite from re-upload is captured in the audit log.	N/E
REG-UPL-05	Upload	P2-High	**Bulk upload with a deactivated lab name is rejected with an appropriate error**	*Upload CSV references 'Beta Lab' which was deactivated*	1. Upload file referencing 'Beta Lab'. 2. Review upload result.	Rows with 'Beta Lab' fail with error: 'Lab Beta Lab is inactive and cannot be used for data entry.' Active lab name required. Error is per-row — other rows with active labs succeed.	N/E
REG-UPL-06	Upload	P2-High	**Alternate Format upload triggers OOR flags identically to Primary Format upload**	*Alternate Format file contains WBC=16.0 for subject S-902; range upper=11.0*	1. Upload via Alternate Format Data Upload. 2. Open CBC Panel for S-902.	OOR flag fires for WBC=16.0 after Alternate Format upload — identical behavior to Primary Format. Upload method does not affect flag evaluation logic.	N/E
REG-UPL-07	Upload	P1-Critical	**Upload mapping does not carry over Lab field**	*Mapping for CBC Panel has Lab field mapped; user*	1. Navigate to Configure → Bulk Upload. 2. Use the CBC Panel mapping	The Lab field mapping is form-specific. A mapping created for	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
			assignment to a different Lab form	*tries to use the same mapping for Metabolic Panel*	for a Metabolic Panel upload (or attempt to select an incompatible mapping).	CBC Panel cannot be applied to Metabolic Panel without creating a new mapping. System prevents cross-form mapping reuse.	
REG-UPL-08	**Upload**	**P2-High**	**Primary Format upload preserves the Lab field value across partial field updates**	*S-903 has Alpha Lab selected on CBC Panel from a previous upload; a new Primary Format upload updates only the WBC Count field*	1. Upload a Primary Format file that only updates WBC Count for S-903 (Lab field not included). 2. Open S-903's CBC Panel.	Alpha Lab selection is preserved. A partial upload that doesn't include the Lab field does not clear or reset the previously set lab selection.	**N/E**
REG-UPL-09	**Upload**	**P1-Critical**	**Bulk upload of 200+ subjects processes all rows and applies flags correctly without timeout**	*200-row bulk upload file for CBC Panel; 50 rows have WBC values outside the reference range*	1. Upload the 200-row file. 2. After completion, navigate to 5 random subjects with flagged WBC values. 3. Verify flags are present.	All 200 rows are processed. The 50 out-of-range rows each have correct higher/lower flags. No timeout or partial processing due to volume. Upload completion is confirmed by the result report.	**N/E**

REG-EXP Export — Column Fidelity & Flag State Regression

Validates all 8 required Include Labs columns, flag state accuracy, and deactivated lab/range data inclusion in exports.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-EXP-01	**Export**	**P2-High**	**Data Export with Include Labs produces exactly 8 Lab-specific columns — no additions, no omissions**	*Hematology Panel has subject data; export includes Labs*	1. Export with Include Labs. 2. Open Excel. 3. Count and list all Lab-specific columns for a Lab form.	Exactly 8 Lab columns: Lab Name, Flag State, Reference Range values, Range Determinant values, Out-of-Range (Within/Above/Below/Equal/Not Equal), Lower Limit, Upper Limit, Expected Value. No duplicates, no missing columns.	**N/E**
REG-EXP-02	**Export**	**P1-Critical**	**Flag State column values are limited to exactly 5 defined statuses — no free text or variants**	*Export includes subjects with all possible flag states*	1. Download export with Include Labs. 2. Open the Out-of-Range column. 3. Inspect all distinct values present.	Out-of-Range column contains only: Within, Above, Below, Equal, Not Equal. No other values (e.g., 'Inactive', 'Triggered', 'Unknown', blank) appear. All 5 statuses are accounted for.	**N/E**
REG-EXP-03	**Export**	**P2-High**	**Export Labs Excel tab count matches the number of all labs (active + deactivated)**	*3 active labs, 2 deactivated labs; all have at least one reference range*	1. Export Labs. 2. Count the number of tabs in the Excel file.	Excel file has 5 tabs (one per lab) plus one lab details overview tab. Deactivated labs have their own tab with Inactive status noted.	**N/E**

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						Total tabs = active + deactivated labs + details tab.	
REG-EXP-04	**Export**	**P2-High**	**Exported range determinant values match what is shown in the Field Information popup**	*S-1000 triggered a flag; popup shows Gender=Female as the matched determinant*	1. Export with Include Labs for S-1000. 2. Open Range Determinant values column.	Range Determinant column for S-1000 shows 'Gender = Female'. Matches exactly what the Field Information popup reports as the matched determinant condition.	**N/E**
REG-EXP-05	**Export**	**P2-High**	**Export correctly shows 'Within' for a subject whose value was corrected from out-of-range to in-range**	*S-1001 had WBC=14.0 (Above); corrected to 7.0 (Within)*	1. Export with Include Labs after the correction. 2. Find S-1001's WBC row.	Out-of-Range column shows 'Within' for the corrected entry. The export reflects the current flag state (Inactive/Within), not the original flagged state.	**N/E**
REG-EXP-06	**Export**	**P2-High**	**LOINC code and unit appear in Export Labs for deactivated reference ranges**	*Deactivated range 'TSH Standard' had LOINC='3016-3' and unit='mIU/L'*	1. Export Labs. 2. Find TSH Standard in the relevant lab tab.	LOINC='3016-3' and unit='mIU/L' are present in the export row for 'TSH Standard', even though the range is deactivated. Metadata is not stripped from deactivated range exports.	**N/E**
REG-EXP-07	**Export**	**P2-High**	**Export includes Lab Name correctly for subjects who used multiple different labs across visits**	*S-1002: Screening = Alpha Lab, Week 4 = Beta Lab, Week 8 = Alpha Lab*	1. Export with Include Labs for Hematology Panel. 2. Filter for S-1002 rows. 3. Check Lab Name per visit row.	Three rows for S-1002: Lab Name = Alpha Lab (Screening), Beta Lab (Week 4), Alpha Lab (Week 8). Each visit row carries the lab that was selected for that specific visit.	**N/E**
REG-EXP-08	**Export**	**P1-Critical**	**Export without Include Labs contains zero Lab Management fields for Lab forms**	*Hematology Panel has Lab data with OOR flags*	1. Export Hematology Panel without checking Include Labs. 2. Open Excel → inspect all column headers.	No Lab Name, Flag State, Out-of-Range, Lower/Upper Limit, Expected Value, or Reference Range columns are present. Export is identical to a non-Lab form export. Include Labs checkbox is the sole gate for lab columns.	**N/E**

REG-AUD Audit Logs — Completeness & Accuracy Regression

🔍 *Validates audit logs capture all change types — lab details, ranges, LOINC, effective dates — with correct user attribution and timestamps.*

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-AUD-01	**Audit**	**P1-Critical**	**Lab Details Audit Log records all 8 column types for a single lab name change**	*Lab 'Alpha Lab' renamed to 'Alpha Clinical Lab' by user qa_architect*	1. Download Lab Details Audit Log. 2. Find the rename entry for Alpha Lab. 3. Verify all 8 columns have data.	Row has: Lab Name=Alpha Clinical Lab, Object=Lab Name, Action=Updated, Old Value=Alpha Lab, New Value=Alpha Clinical Lab, Date=(correct date), Time=(correct time), Changed By=qa_architect. All 8 columns populated, none blank.	**N/E**
REG-AUD-02	**Audit**	**P1-Critical**	**Lab Details Audit Log records lab code addition as a separate entry from lab name changes**	*Lab code 'ALP-001' was added after the lab name was already saved*	1. Add lab code 'ALP-001' to Alpha Lab → save. 2. Download Lab Details Audit Log → filter Alpha Lab.	Two separate audit entries: one for Lab Name (from earlier) and one for Lab Code (Object=Lab Code, Action=Added, New Value=ALP-001). Distinct fields generate distinct audit rows.	**N/E**
REG-AUD-03	**Audit**	**P1-Critical**	**Reference Range Audit Log captures a lower limit change with exact old and new values**	*WBC Adult Range lower limit changed from 4.5 to 4.0 by user qa_lead*	1. Change lower limit → save. 2. Download Reference Range Audit Log → find the entry.	Row: Lab Form=CBC Panel, Lab Test Name=WBC Count, Range Name=WBC Adult Range, Object=Lower Limit, Action=Updated, Old Value=4.5, New Value=4.0, Changed By=qa_lead. Values are exact (not rounded).	**N/E**
REG-AUD-04	**Audit**	**P1-Critical**	**Reference Range Audit Log captures Range Determinant addition with the variable name and value**	*Gender=Female added as Range Determinant to 'WBC Female Range' by qa_manager*	1. Add Gender=Female determinant → save. 2. Download Reference Range Audit Log → find the determinant entry.	Row: Object=Range Determinant, Action=Added, New Value='Gender = Female', Changed By=qa_manager. Old Value is blank (no previous determinant). Range Name and Lab Form are correctly attributed.	**N/E**
REG-AUD-05	**Audit**	**P1-Critical**	**Both audit logs capture the correct username even for shared service accounts**	*A shared service account 'svc_upload' performs a lab detail change and a range value change*	1. Log in as svc_upload → update a lab address and a range limit. 2. Download both audit logs.	Both audit logs show Changed By = 'svc_upload'. The system uses the authenticated username, not a generic system or admin label, for audit attribution.	**N/E**
REG-AUD-06	**Audit**	**P1-Critical**	**Lab Details Audit Log captures lab reactivation as a separate row from deactivation**	*Beta Lab was deactivated and then reactivated in the same study week*	1. Download Lab Details Audit Log → filter Beta Lab.	Two separate rows: Row 1: Action=Deactivated, Old Value=Active, New Value=Inactive. Row 2: Action=Reactivated, Old Value=Inactive, New Value=Active. Timestamps are sequential and distinct. Both rows attribute the correct users.	**N/E**

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-AUD-07	Audit	P2-High	**Reference Range Audit Log entries for bulk effective dates are time-stamped within 30 seconds of each other**	*5 ranges received bulk effective dates in a single Apply action*	1. Apply bulk dates to 5 ranges. 2. Download Reference Range Audit Log. 3. Check timestamps for all 10 entries (2 per range).	All 10 audit entries (5 Start Date + 5 End Date) have timestamps within 30 seconds of each other. They are clearly clustered as a single batch action, not spread across minutes.	N/E
REG-AUD-08	Audit	P2-High	**Audit log file is non-empty and opens without error even when no changes were made in the current session**	*Audit log is downloaded immediately after logging in without making any changes*	1. Log in → navigate to Lab Management. 2. Click three-dot → Lab Details Audit Log → download. 3. Open the file.	Excel file downloads and opens without error. It contains historical entries from previous sessions. The file is not blank or corrupt. Historical audit data is always available.	N/E
REG-AUD-09	Audit	P1-Critical	**Reference Range Audit Log is not affected by lab details changes — only range-level changes appear**	*Lab address and name were updated; separately, a range limit was changed*	1. Download Reference Range Audit Log. 2. Search for lab address or name change entries.	Reference Range Audit Log contains only range-level changes (limits, determinants, effective dates, LOINC, status). Lab name and address changes do NOT appear in the Reference Range Audit Log — they belong only in the Lab Details Audit Log.	N/E
REG-AUD-10	Audit	P2-High	**Deactivated reference range's Deactivation entry appears in audit but no further entries are added after deactivation**	*'WBC Pediatric Range' was deactivated; no one edits it after deactivation*	1. Deactivate 'WBC Pediatric Range'. 2. Download Reference Range Audit Log → find all entries for this range. 3. Verify no post-deactivation modification entries exist.	Audit contains exactly one deactivation entry (Action=Deactivated). No subsequent entries exist after the deactivation timestamp. The range is read-only post-deactivation — no further audit rows are generated.	N/E

REG-DRL Deactivate/Reactivate Lab — Flag & Data Persistence Regression

Confirms flag persistence post-deactivation, configuration integrity post-reactivation, and export inclusion of deactivated labs.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-DRL-01	Deact/React	P2-High	**Deactivating a lab removes it from the data entry dropdown but retains its reference ranges in Lab Management**	*'Gamma Lab' is deactivated; previously had 5 reference ranges*	1. Deactivate Gamma Lab. 2. Open a Lab form for any subject → check Lab dropdown. 3. Navigate to Manage → Lab Management → Deactivated Labs → Gamma Lab → LAB TESTS → Manage Ranges.	Lab dropdown: Gamma Lab is absent. Lab Management Deactivated Labs: Gamma Lab's 5 reference ranges are visible (read-only). Configuration is preserved for audit/reference purposes.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-DRL-02	Deact/React	P1-Critical	**Historical OOR flags remain in Data Export for subjects who used a deactivated lab**	*20 subjects used Gamma Lab; Gamma Lab is now deactivated*	1. Export data with Include Labs for the relevant Lab form. 2. Filter by Lab Name = 'Gamma Lab'.	20 subject rows appear with Lab Name = 'Gamma Lab' and correct Out-of-Range statuses. Deactivating a lab does not purge or anonymize historical lab-linked data from exports.	N/E
REG-DRL-03	Deact/React	P2-High	**Reactivating a lab restores it to the data entry dropdown immediately**	*Gamma Lab was deactivated and is being reactivated*	1. Navigate to Deactivated Labs → Reactivate Gamma Lab. 2. Open any Lab form for a subject → click Lab field.	Gamma Lab reappears in the data entry Lab dropdown immediately after reactivation. No cache clear or session restart is required.	N/E
REG-DRL-04	Deact/React	P2-High	**Reactivated lab's reference ranges re-evaluate for new data entries**	*Gamma Lab is reactivated; WBC range 4.5–11.0 is restored*	1. Open a Lab form → select Gamma Lab → enter WBC=14.0 → save.	Higher OOR flag triggers for WBC=14.0 under Gamma Lab (upper limit 11.0). Range evaluation is fully functional after reactivation.	N/E
REG-DRL-05	Deact/React	P1-Critical	**Lab deactivation event is captured in the Lab Details Audit Log**	*Gamma Lab was deactivated by user qa_manager*	1. Download Lab Details Audit Log → filter Gamma Lab.	Audit row: Object=Lab Status, Action=Deactivated, Old Value=Active, New Value=Inactive, Changed By=qa_manager. Date and time are accurate.	N/E
REG-DRL-06	Deact/React	P2-High	**Deactivated lab count in Deactivated Labs list is accurate**	*3 labs have been deactivated total in the study*	1. Navigate to Manage → Lab Management → three-dot → Deactivated Labs.	Deactivated Labs screen shows exactly 3 labs. Each deactivated lab's DETAILS, selected forms, and reference ranges are visible.	N/E
REG-DRL-07	Deact/React	P2-High	**Lab cannot be deactivated by a user who has only Add/Edit Labs permission and not Deactivate Labs**	*User has Add/Edit Labs only; no Deactivate Labs permission*	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to an active lab. 3. Check if 'Deactivate Lab' option appears.	'Deactivate Lab' option is absent from the three-dot menu for this user. Only users with explicit Deactivate Labs permission can see and use the deactivation option.	N/E
REG-DRL-08	Deact/React	P2-High	**A newly added lab after mid-study deactivations does not inherit any historical configuration**	*3 labs were deactivated; a new 'Delta Lab' is created via the + icon*	1. Add Delta Lab via + icon. 2. Navigate to Delta Lab → DETAILS tab → LAB TESTS tab.	Delta Lab has empty DETAILS fields (no code, no address pre-populated). LAB TESTS tab has no pre-selected forms. It is a completely fresh lab with no inherited configuration from deactivated labs.	N/E

REG-MIG Study Migration — Config Lock & Permission Regression

🔍 *Validates configuration lock after deployment, permission propagation, and correct environment scoping of labs/ranges.*

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-MIG-01	**Migration**	**P1-Critical**	**Lab Test Result setting is locked in TEST after DEV → TEST deployment**	*WBC Count is Lab Test Result in DEV; deployed to TEST*	1. In TEST, navigate to Configure → Forms → CBC Panel → WBC Count → Variable Editor. 2. Attempt to change Lab setting from 'Lab Test Result' to 'None'.	Setting change is blocked in TEST: 'This variable has been deployed to a child study or used in Lab Management. Settings cannot be changed.' DEV-configured lab variable settings are immutable in child environments.	**N/E**
REG-MIG-02	**Migration**	**P1-Critical**	**Range Determinant variable cannot be reconfigured in TEST post-deployment**	*Gender is Range Determinant in DEV; deployed to TEST*	1. In TEST, open Gender Variable Editor. 2. Attempt to change Lab Range setting from 'Range Determinant' to 'None'.	System blocks the change: 'Configuration deployed from parent study cannot be modified.' Range Determinant setting is locked in TEST.	**N/E**
REG-MIG-03	**Migration**	**P1-Critical**	**Labs and reference ranges created in TEST do not deploy to LIVE**	*TEST study has 'TEST_Lab_A' created for testing; deploying TEST → LIVE*	1. In TEST, create 'TEST_Lab_A' with a reference range. 2. Deploy TEST config to LIVE. 3. In LIVE, navigate to Manage → Lab Management.	'TEST_Lab_A' does NOT appear in LIVE. Labs and reference ranges are manually managed in each environment and are not part of the configuration deployment package.	**N/E**
REG-MIG-04	**Migration**	**P1-Critical**	**Lab permissions for roles are preserved across DEV → TEST → LIVE deployments**	*QA_Role has Add/Edit Labs in DEV; deployed to TEST and then LIVE*	1. In TEST → Configure → Roles & Permissions → QA_Role → PERMISSIONS tab. 2. In LIVE → same navigation.	Both TEST and LIVE retain Add/Edit Labs for QA_Role. Permission configuration is part of the deployment package and is preserved. No manual re-granting of permissions needed.	**N/E**
REG-MIG-05	**Migration**	**P2-High**	**Form visit assignments for Lab forms are preserved in TEST after DEV deployment**	*CBC Panel in DEV is assigned to Screening, Week 4, Week 8*	1. Deploy DEV to TEST. 2. In TEST, navigate to Configure → Forms → CBC Panel → check visit assignments.	All three visits (Screening, Week 4, Week 8) are associated with CBC Panel in TEST. Visit assignments are part of the form configuration and survive deployment.	**N/E**
REG-MIG-06	**Migration**	**P1-Critical**	**LIVE study allows adding new labs and ranges not present in DEV/TEST**	*LIVE study has CBC Panel as Lab form (deployed); no labs created in LIVE yet*	1. In LIVE, navigate to Manage → Lab Management. 2. Create 'Production Lab' → add WBC reference range 4.5–11.0. 3. Test data entry with 'Production Lab'.	LIVE study allows creating 'Production Lab' and configuring ranges independently. Labs/ranges are LIVE-specific and not locked by deployment. Data entry users in LIVE can use Production Lab.	**N/E**

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-MIG-07	Migration	P2-High	**DEV study can still create and modify labs even after deploying to TEST**	*DEV has been deployed to TEST; Dev team is continuing DEV configuration*	1. In DEV, navigate to Manage → Lab Management. 2. Create a new lab 'DEV_Lab_Temp'. 3. Modify an existing reference range.	DEV study allows full lab and range management after deployment. Deployment does not lock the DEV environment. Changes in DEV can be deployed again to TEST after further testing.	N/E
REG-MIG-08	Migration	P1-Critical	**Lab form use type change after deployment attempt is blocked in TEST**	*CBC Panel is deployed to TEST as a Lab form*	1. In TEST, navigate to Configure → Forms → CBC Panel. 2. Attempt to change form use type from 'Lab' to '--N/A--'.	System blocks the change: 'This form has been deployed from a parent study. Form use type cannot be modified.' CBC Panel remains as a Lab form in TEST.	N/E
REG-MIG-09	Migration	P2-High	**EDC Admin Labs flag state is preserved after deployment**	*Labs is enabled under EDC Admin in DEV before deployment*	1. Deploy DEV to TEST. 2. In TEST, navigate to EDC Admin → Enable → check Labs checkbox state.	Labs checkbox remains enabled in TEST after deployment. EDC Admin configuration is part of the deployment and is preserved. Lab Management is accessible in TEST without re-enabling.	N/E

REG-NEG Negative Paths — Unsupported Feature Guards

Confirms that unsupported features are consistently blocked across system updates and that boundary violations are properly rejected.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-NEG-01	Negative	P2-High	**DDE is blocked for Lab forms after a system update that touched DDE logic**	*A system update modified Double Data Entry workflows; Lab form 'CBC Panel' exists*	1. Attempt to configure or initiate DDE for CBC Panel. 2. Observe system behavior after the update.	DDE is still not available for Lab forms. System update did not inadvertently enable DDE for Lab forms. Restriction is enforced identically post-update.	N/E
REG-NEG-02	Negative	P2-High	**MSDE is blocked for Lab forms after a system patch**	*Multiple Subject Data Entry was modified in a recent patch*	1. Attempt to use MSDE for Hematology Panel Lab form. 2. Observe system behavior.	MSDE remains unavailable for Lab forms. The patch did not enable MSDE for Lab forms. Restriction is consistent.	N/E
REG-NEG-03	Negative	P2-High	**Real-time reports do not include Lab Management flag data after a reporting engine update**	*Real-time reports feature was updated; Lab forms exist with flag data*	1. Navigate to real-time reports. 2. Attempt to include Lab form flag columns (Flag State, Reference Range, Lab Name).	Lab Management-specific columns are absent from real-time report options. Reporting engine update did not inadvertently expose lab flag data in real-time reports.	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-NEG-04	Negative	P1-Critical	**Import API returns an error when a Lab field value is included in the payload**	*Import API is active; payload includes a Lab form field submission with a lab selection*	1. Submit an API payload for a Lab form that includes a Lab field value. 2. Review the API response.	API returns an error indicating Lab Management data is not supported via Import API. The Lab field value is not processed. System does not silently ignore the lab field — it explicitly rejects it.	N/E
REG-NEG-05	Negative	P1-Critical	**Export API response does not include out-of-range flag state or reference range columns**	*Export API is called for a study with Lab forms and active OOR flags*	1. Call Export API for CBC Panel. 2. Parse the response for Flag State, Out-of-Range, Reference Range, Lab Name columns.	API response contains no lab-specific columns. Flag State, Out-of-Range, Reference Range, Lower/Upper Limit, Expected Value are all absent from the API export. These are only available via the UI Data Export with Include Labs.	N/E
REG-NEG-06	Negative	P2-High	**BYO reports do not expose Lab Management-specific fields for Lab forms**	*BYO report builder is open; CBC Panel is available as a data source*	1. Open BYO report builder. 2. Add CBC Panel as the form. 3. Browse available fields for lab-specific data (Lab Name, Flag State, OOR status).	Lab Name, Flag State, Reference Range, Out-of-Range, and similar Lab Management-specific fields are not available in the BYO report field picker for Lab forms.	N/E
REG-NEG-07	Negative	P1-Critical	**Submitting a Lab form with WBC Count blank (null) does not create a ghost out-of-range flag**	*Alpha Lab range for WBC Count: 4.5–11.0; form is saved with empty WBC field*	1. Select Alpha Lab → leave WBC Count blank → save. 2. Check for any OOR flag next to WBC Count.	No flag appears. Null/blank values are excluded from range comparison. The range is not evaluated when the field has no value. No ghost flag is created.	N/E
REG-NEG-08	Negative	P2-High	**Entering a non-numeric value in a numeric Lab Test Result variable is blocked before OOR evaluation**	*WBC Count is a numeric Lab Test Result variable; user attempts to enter 'N/A'*	1. Open CBC Panel → select a lab. 2. Type 'N/A' in WBC Count field. 3. Attempt to save.	System blocks save with: 'Invalid entry — numeric value required.' OOR evaluation is never reached for non-numeric input. The type validation fires before range comparison.	N/E
REG-NEG-09	Negative	P2-High	**Accessing Lab Management URL directly by a non-Lab user in LIVE is blocked**	*A user in LIVE has no Lab permissions; knows the Lab Management URL*	1. Log in as non-Lab user in LIVE. 2. Navigate to the Lab Management URL directly.	Access is denied. System redirects to the dashboard or shows a permission error. Production environment (LIVE) enforces the same permission checks as DEV/TEST.	N/E
REG-NEG-10	Negative	P2-High	**Creating a lab with a blank name is blocked with a validation error**	*User has Add/Edit Labs permission; Create Labs form is open*	1. Leave the lab name field blank in the Create Labs form. 2. Click Create.	System blocks the creation: 'Lab name cannot be blank.' No empty-named lab is created. The Create Labs form requires at least one valid lab name.	N/E

REG-PER Performance & Concurrent Session Regression

Validates system behavior under production-representative data volumes and concurrent user sessions.

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
REG-PER-01	Performance	P3-Medium	**Lab Management page loads within 5 seconds when 50+ active labs exist**	*Study has 50 active labs, each with 5–10 reference ranges*	1. Navigate to Manage → Lab Management. 2. Start timing from click to full page render (all labs listed).	Lab Management page renders within 5 seconds. Lab list, Labs (#) counter, DETAILS and LAB TESTS tabs, and three-dot icons all load completely. No timeout or blank page.	N/E
REG-PER-02	Performance	P3-Medium	**Manage Ranges loads all reference ranges within 5 seconds for a lab with 20+ ranges**	*A single lab has 20 reference ranges across 3 Lab forms*	1. Select the lab → LAB TESTS → check all 3 forms → Manage Ranges. 2. Time the Manage Ranges page load.	Manage Ranges displays all 20 ranges (across 3 forms) within 5 seconds. Range names, edit icons, and effective date Calendar icons are all rendered. No ranges are hidden by pagination.	N/E
REG-PER-03	Performance	P3-Medium	**Export Labs Excel file generates and downloads within 60 seconds for a study with 30 labs**	*Study has 30 labs, each with average 8 reference ranges = 240 total ranges*	1. Click Export Labs. 2. Time from click to file download completion.	Excel file download initiates within 60 seconds. File contains all 30 lab tabs plus a details tab. File opens without corruption. Performance degrades gracefully — no server error for high-volume exports.	N/E
REG-PER-04	Performance	P3-Medium	**Concurrent reference range saves by two different admins do not corrupt each other's data**	*Admin A edits WBC lower limit; Admin B edits Hemoglobin upper limit simultaneously*	1. Admin A edits WBC Adult Range lower limit to 4.0 and saves. 2. Admin B simultaneously edits Hemoglobin Standard upper limit to 17.0 and saves. 3. Reload both ranges.	WBC Adult Range lower = 4.0 (Admin A's change). Hemoglobin Standard upper = 17.0 (Admin B's change). No cross-contamination. Each admin's save is isolated to their respective range.	N/E
REG-PER-05	Performance	P3-Medium	**Data Export with Include Labs completes within 120 seconds for 1000+ subject records**	*Study has 1000 subjects with Hematology Panel data; 80% have OOR flags*	1. Navigate to Analysis → Data Export → Hematology Panel → Include Labs → Download. 2. Time from click to download completion.	Export file downloads within 120 seconds. All 1000 subject rows are present in the file with correct Lab Name, Out-of-Range, Flag State columns. No timeout or partial export.	N/E
REG-PER-06	Performance	P3-Medium	**Lab Details Audit Log download is performant for a study with 2+ years of lab change history**	*Study has been active for 2 years; Lab Details Audit Log has 5000+ entries*	1. Click three-dot → Lab Details Audit Log → download. 2. Open the file.	Excel file downloads within 30 seconds. File opens without hanging or crashing Excel. All 5000+ rows are present with	N/E

REG-ID	Area	Priority	Regression Test Title	Pre-Conditions	Test Steps (Real-Time)	Expected Result	Status
						correct column data. File is not truncated.	

Regression Test Execution Sign-Off

Role	Full Name	Signature / Approval	Date
QA Manager / Senior QA Architect			
QA Lead			
Development Lead			
Product Owner			
Regulatory Affairs			